Please find below the preliminary term project proposal for the DA592 Summer 2026 term.

1. Project Title

RoamWise: An Agentic AI Framework for Personalized Tourism Forecasting and Intelligent Itinerary Optimization

2. Team Members

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3. Motivation

The modern travel industry is transitioning from static, one-size-fits-all tour packages to hyper-personalized, data-driven experiences. Currently, travelers struggle to align their unique preferences (budget, cultural interests) with macro-level tourism constraints such as seasonal demand spikes, crowding, and dynamic pricing. Conversely, identifying optimal geographic routes in unfamiliar cities remains a complex operational challenge.

Critically, travel and destination data is inherently relational: destinations are connected to transport hubs, transport hubs are linked to time-constrained activity clusters, and user preferences intersect with all of these dimensions simultaneously. Standard retrieval approaches that treat these as independent documents fail to capture this web of relationships. This makes Graph-RAG a particularly critical architectural component for this domain, as it enables the system to map destinations, transport options, and user preferences not as isolated facts but as a richly interconnected network.

This problem presents an excellent opportunity to explore how isolated analytical models—specifically time-series forecasting and optimization—can be autonomously orchestrated by Large Language Models (LLMs). The motivation is to build a scalable pipeline where GenAI acts not just as a conversational interface, but as a central reasoning engine that invokes predictive and optimization tools and leverages a Fusion RAG architecture (Semantic + Graph + Keyword) to solve multi-hop travel planning queries with precision.

4. Project Description

The project aims to develop "RoamWise," an intelligent travel planning ecosystem driven by an Agentic Flow architecture. The system will address the challenge of destination selection and daily routing by solving three interconnected sub-problems:

- Predicting macro-level tourism trends to recommend destinations that avoid peak crowding or exploit off-peak advantages.
- Segmenting users based on travel constraints and preferences to filter relevant points of interest (POIs).
- Generating geographically optimized, day-by-day itineraries using algorithmic routing constraints, enriched by graph-traversal outputs.

Ultimately, the goal is to observe how effectively an LLM can synthesize predictive data, knowledge graph traversal results, unstructured cultural texts, and geographical constraints to generate highly coherent, actionable outputs.

5. Methodology

To maintain architectural flexibility, the project will be developed using a modular, multi-agent framework rather than a monolithic pipeline. The methodology consists of the following components:

- **Data Sourcing:** Utilizing open-source aviation, tourism demand, and POI datasets (e.g., Kaggle, TripAdvisor, OpenStreetMap, Wikidata) to ensure a robust foundation without confidentiality constraints. Structured relational data will additionally populate the knowledge graph.

- Predictive & Segmentation Models (The Tools):

– Demand Forecasting: Implementing time-series models (e.g., Prophet, LSTM) to predict destination popularity and passenger volumes.

– Traveler & POI Segmentation: Utilizing unsupervised learning algorithms (clustering) to segment user profiles at the input stage, and later to spatially group POIs into daily geographical zones.

- **Fusion RAG Architecture (Core Retrieval Layer):** As recommended, the retrieval layer is designed as a three-component Fusion RAG system:
  - Semantic Search: Dense vector retrieval (e.g., FAISS, ChromaDB) over unstructured city guides for contextually similar passages.
  - Graph-RAG (Knowledge Graph Traversal): A structured knowledge graph maps destinations, transport options, POIs, activity chains, and user preference nodes as interconnected entities. Graph traversal (e.g., via Neo4j or NetworkX) enables the system to resolve complex, multi-hop queries—such as chaining transport links or sequencing constraint-aware activities—that flat document retrieval cannot handle.
  - Keyword Search (BM25): Sparse lexical retrieval to capture exact-match entities (destination names, landmark titles, transport codes) that semantic embeddings may underweight.
  - The three signals are fused using reciprocal rank fusion (RRF) or a learned re-ranking mechanism before being passed to the LLM agents as grounded context.

- **Agentic Orchestration (Core Focus):** Implementing an LLM orchestration framework (e.g., LangGraph or similar multi-agent systems) featuring specialized agents:
  - Forecaster Agent: Interprets the outputs of the time-series models.
  - Fusion RAG Agent: Orchestrates the three-component retrieval pipeline, prioritizing Graph-RAG traversal for multi-hop relational queries and falling back to semantic/keyword search for descriptive content retrieval.
  - Router Agent: Acts as an algorithmic orchestrator, utilizing an optimization methodology as a tool to solve the routing problem logically—informed by graph-enriched context—before generating the final text output.

- **Comparative Analysis:** As recommended, the project will document a structured comparative analysis across three retrieval configurations:
  - Fusion RAG (Semantic + Graph + Keyword) — primary architecture
  - Hybrid RAG (Semantic + Keyword only) — ablation baseline
  - Standard prompting (no retrieval) — lower-bound reference

Evaluation will focus on geographical hallucination rate, multi-hop query resolution accuracy, and itinerary coherence.

6. Project Deliverables / Expected Results

- **An Interactive Prototype:** A web-based UI (e.g., Streamlit) demonstrating the end-to-end flow, allowing a user to input preferences and receive a data-backed, graph-grounded, optimized itinerary.
- **Agentic Architecture Codebase:** The underlying modular code integrating ML models, the knowledge graph, and the Fusion RAG pipeline with the LLM orchestrator.
- **Knowledge Graph:** A structured graph database linking destinations, transport options, POIs, and user preference profiles.
- **Final Technical Report:** A comprehensive document detailing the methodology, the comparative analysis of retrieval architectures (Fusion RAG vs. Hybrid RAG vs. standard prompting), and an evaluation of the LLM's ability to perform relational reasoning for logistical routing.

7. Milestones

The project will follow a modular timeline to allow iterative development and testing:

- **Weeks 1–2 (Data & Baseline Models):** Dataset acquisition, exploratory data analysis (EDA), and development of baseline time-series and clustering algorithms.
- **Weeks 3–4 (Knowledge Base & Fusion RAG Construction):** Preparation of unstructured text data, embedding generation, knowledge graph construction (destinations, POIs, transport links, user preferences), and setup of the Fusion RAG pipeline (Semantic + Graph + Keyword).
- **Weeks 5–7 (Agentic Flow Integration):** Development of the multi-agent orchestration layer; integration of Graph-RAG traversal for multi-hop reasoning; defining tools (routing heuristics, forecasting models, graph queries) for the LLM agents to utilize.
- **Weeks 8–9 (UI Development & Refinement):** Building the interactive frontend interface and tuning system prompts to reduce hallucinations and improve coherence of graph-enriched outputs.
- **Week 10 (Evaluation & Documentation):** Final end-to-end testing, compiling the comparative analysis (Fusion RAG vs. Hybrid RAG vs. standard prompting), and finalizing the project report.

Project Phases / Weeks	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10
1. Data Sourcing & Baseline Models	■	■								
2. Knowledge Base & RAG Setup			■	■						
3. Agentic Flow Orchestration					■	■	■			
4. UI Development & Prompt Tuning								■	■	
5. Evaluation & Final Report										■